

Program 5: 2D Matrix Subtraction

Aim

To write a C program to subtract one matrix from another.

Theory

A **matrix** is a two-dimensional array of numbers arranged in rows and columns.

Two matrices can be subtracted only if they have the **same order**.

The difference $C = A - B$ is obtained by subtracting the corresponding elements: $C[i][j] = A[i][j] - B[i][j]$.

Unlike addition, subtraction is **not commutative**: $A - B$ is generally not equal to $B - A$. Time complexity: $O(r \times c)$.

Algorithm

1. Read the number of rows r and columns c .
2. Read the elements of matrices A and B .
3. For each i from 0 to $r-1$ and each j from 0 to $c-1$, compute $C[i][j] = A[i][j] - B[i][j]$.
4. Display matrix C .

C Source Code

```
/*
 * Program : 2D Matrix Subtraction
 *
 * THEORY:
 * - Two matrices can be subtracted only if they have the SAME order (rows x cols).
 * -  $C[i][j] = A[i][j] - B[i][j]$  for every position  $(i, j)$ .
 * - Subtraction is NOT commutative ( $A-B \neq B-A$  in general).
 * - Time complexity:  $O(r * c)$ .
 */
#include <stdio.h>
#define MAX 10

void readMatrix(int m[MAX][MAX], int r, int c, char name) {
    printf("Enter elements of matrix %c (%d x %d):\n", name, r, c);
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            scanf("%d", &m[i][j]);
}

void printMatrix(int m[MAX][MAX], int r, int c) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++)
            printf("%5d", m[i][j]);
        printf("\n");
    }
}

int main() {
    int A[MAX][MAX], B[MAX][MAX], C[MAX][MAX], r, c;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &r, &c);

    readMatrix(A, r, c, 'A');
    readMatrix(B, r, c, 'B');

    for (int i = 0; i < r; i++)
```

```

        for (int j = 0; j < c; j++)
            C[i][j] = A[i][j] - B[i][j];

    printf("\nMatrix A:\n"); printMatrix(A, r, c);
    printf("\nMatrix B:\n"); printMatrix(B, r, c);
    printf("\nResult (A - B):\n"); printMatrix(C, r, c);
    return 0;
}

```

Sample Input

Input: 2 x 3 matrices A = [1 2 3; 4 5 6] and B = [9 8 7; 6 5 4].

Sample Output (program run with the input above)

```

Enter number of rows and columns: Enter elements of matrix A (2 x 3):
Enter elements of matrix B (2 x 3):

```

Matrix A:

```

1    2    3
4    5    6

```

Matrix B:

```

9    8    7
6    5    4

```

Result (A - B):

```

-8   -6   -4
-2    0    2

```